PARUL UNIVERSITY

Faculty of Engineering & Technology
B.Tech – Semester III
Subject: Design of Data Structures (303105201)

Time: 2.5 Hours
Max Marks: 60

SECTION – A (30 Marks)

Q1 (Compulsory) – 12 Marks

a) Answer the following (ANY THREE): ($3 \times 2 = 6$)
1. Define Data Structure.
2. What is Stack?
3. Define Queue.
4. What is a Pointer?

b) Answer the following (ANY THREE): ($3 \times 2 = 6$)
1. Define Binary Tree.
2. What is Hashing?
3. Define Graph.
4. What is Recursion?

Q2 – 9 Marks

a) Explain Stack operations with diagram. (4)

b) Answer ANY ONE: (5)

- Explain Infix to Postfix conversion.
- Evaluate postfix expression: `5 3 + 8 2 - *`

Q3 – 9 Marks

a) Explain Linear Search algorithm. (4)

b) Answer ANY ONE: (5)

- Explain Binary Search with example.

- Explain Interpolation Search.

SECTION – B (30 Marks)

Q4 (Compulsory) – 12 Marks

a) Answer the following (ANY THREE): ($3 \times 2 = 6$)
1. Define Linked List.
2. What is Circular Queue?
3. Define AVL Tree.
4. What is BFS?

b) Answer the following (ANY THREE): ($3 \times 2 = 6$)
1. Define BST.
2. What is Collision?
3. Define DFS.
4. What is Deque?

Q5 – 9 Marks

a) Explain Singly Linked List insertion. (4)

b) Answer ANY ONE: (5)

- Explain Doubly Linked List with diagram.
- Explain Circular Linked List.

Q6 – 9 Marks

a) Explain Binary Tree traversals. (4)

b) Answer ANY ONE: (5)

- Explain Binary Search Tree.
- Explain Expression Tree.